Jinmyeong Kim

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Research Interests

Systems for Embodied Interaction; XR/Mobile Systems

Education

Integrated MS/PhD, Computer Science and Engineering, Seoul National University — Present
Advisor: Youngki Lee

BS, Electrical and Computer Engineering, Seoul National University — Aug 2024

Publications

[1] **Jinmyeong Kim***, Juheon Yi*, Wootack Kim, Seokgyeong Shin, Youngki Lee. Combinational Point Sampling for Fast and Accurate On-Device LiDAR 3D Object Detection. IEEE INFOCOM 2025. (*Equal contribution)

[2] Jingyu Lee, Dongho Han, **Jinmyeong Kim**, Youngki Lee. SpatialBlend: Relational Reconstruction for Cross-Space Presence in Heterogeneous Spaces. Under Review at UIST 2026.

[3] Seokgyeong Shin, Sangwon Park, **Jinmyeong Kim**, Youngki Lee. Failure-Aware Monitoring Code Synthesis for Robotics via Failure Exploration and Validation in Simulation. Under Review at NeurIPS 2026.

Research Experiences

Graduate Researcher, Human-Centered Computer Systems Lab — 2024 – Present
Seoul National University, Seoul, Korea

- Leading an ongoing project on interaction-constrained 6-DoF object pose tracking, leveraging affordance-based interaction priors from the manipulating body to validate and refine neural pose estimates.
- Designed a combinational point-sampling scheme [1] enabling LiDAR 3D object detection with 40% reduction in latency and 35% decrease in energy consumption without accuracy drop on Jetson edge devices.
- Built SpatialBlend's runtime framework [2] — integrating embodiment signals (user 6DoF, pose, voice) that drive a remote-avatar motion backend adapting to reconstructed cross-space behaviors — and conducted the user study using Meta Quest 3.

Visiting Researcher — Aug 2025 – Oct 2025
Purdue University, West Lafayette, IN, USA

- Formulated the research problem of interaction-constrained pose perception during hand manipulation — the foundation of my current ongoing project — through collaboration with Prof. Younghyun Kim's group.
- Initiated a collaboration with the Purdue Spatial Computing Hub on a high-fidelity teleinteraction platform across remote physical spaces, scoping a joint testbed for future evaluation.

Technical Skills

Languages: Python, C++, C#, CUDA, Swift
Frameworks: PyTorch, Qualcomm Hexagon SDK, Unity/OpenXR, Meta XR SDK, Bazel
Hardware: NVIDIA Jetson, Meta Quest 3, Vision Pro

Teaching

Head TA, Mobile Computing and Its Applications, Seoul National University, Spring 2026
Head TA, Advanced Course for AI Scientist - Machine Learning, LG Electronics, Winter 2025

Honors and Awards

Graduated Cum Laude, Seoul National University — 2024
Kim Chung-Shik Scholarship, HaeDong Scholarship Foundation — 2022
Qualcomm IT Tour Mentorship Program, Qualcomm Korea — 2022